2024学年第一学期徐汇区学习能力诊断卷

高三英语　　试卷

（考试时间105分钟　满分115分）　　2024.12

I. Grammar and vocabulary

Section A

Directions: After reading the passage below, fill in the blanks to make the passage coherent and grammatically correct. For the blanks with a given word, fill in each blank with the proper form of the given word; for the other blanks, use one word that best fits each blank.

Pioneers in artificial intelligence win the Nobel Prize in physics

The 2024 Nobel Prize in physics has been awarded to John Hopfield and Geoffrey Hinton. They are known for their fundamental discoveries in machine learning, __1__ which paved the way for how artificial intelligence is used today.

Machine learning differs from traditional software. The software receives data, which is processed according to a clear description, and __2__ producing (produce) the results. In machine learning, the computer learns by example, enabling it to tackle problems that are too complicated __3__ to manage (manage) by step-by-step instructions.

Hinton and Hopfield __4__ be credited (credit) with using tools from physics to advance basic research in the field. In 1982, Hopfield developed a model of *neural* (神经的) networks, today known as the Hopfield network, to describe how the brain recalls memories when __5__ fed (feed) partial information, similar to the method your brain uses to remember a word on the tip of your tongue.

Geoff Hinton and colleagues further developed the Hopfield network. To do that, Hinton used statistical physics, based on an equation invented by the nineteenth century physicist Ludwig Boltzmann, creating a "Boltzmann machine." It can learn—not from instructions, but from __6__ given (give) examples. A trained Boltzmann machine can recognize familiar *traits* (特质) in information it has not previously seen. Imagine meeting a friend's brother or sister, and you can immediately see that they __7__ must be related. In a similar way, the Boltzmann machine can recognize an entirely new example __8__ if what it belongs to a category found in the training material.

Hinton has also urged caution around the technology. Hinton quit his job as a vice president last year at a tech giant. He said he left because he wanted to be able to share his concerns about the risks of artificial intelligence without worrying __9__ what it would mean for his employer.

"One of the ways in which these systems might escape control is by writing their own computer code to modify __10__," Hinton said in a 2023 interview. "That's something we need to seriously worry about."

themselves

BIAKH DCFEJ

Section B

Directions: Fill in each blank with a proper word chosen from the box. Each word can be used only once. Note that there is one word more than you need.

A. present	B. accomplishment	C. plug	D. household	E. defeated
F. significant	G. diagnose	H. addicts	I. amateur	J. alerted	K. picture

A hacker's life

Have you ever locked yourself out of your home and had to try to break in? First, you get a sense of __11__ in succeeding. But then comes the worrying realization that if you can break into your own place as a(n) __12__, a professional could do so five times faster. So you look at the weak point in your security and fix it. Well, that's more or less how the DefCon hackers conference works.

Every year passionate hackers meet at DefCon in Las Vegas to __13__ their knowledge and capabilities. Mention the word 'hacker' and many of us __14__ a seventeen-year-old *geek* (怪人) sitting in their bedroom, illegally hacking into the US's defence secrets in the *Pentagon* (五角大楼). Or we just think 'criminals'. But that is actually a misrepresentation of what most hackers do.

The activities and experiments that take place at DefCon have an enormous impact on our daily lives. These are people who love the challenge of finding security gaps: computer __15__ who can't break the habit. They look with great care at all kinds of systems, from the Internet to mobile communications to __16__ door locks. And then they try to hack them. In doing so, they are doing all of us a great service, because they pass on their findings to the industries that design these systems, which are then able to __17__ the security holes.

A typical example of this is when I attended a presentation on electronic door locks. Ironically, one of the most secure locks they demonstrated was a 4,000-year-old Egyptian lock. But when it came to more modern devices, the presenters revealed __18__ weaknesses in several brands of electro-mechanical locks. A bio-lock that uses a fingerprint scan for entry was __19__, easily, by a paper clip. (Unfortunately, although all the manufacturers of the insecure locks were __20__, not all of them responded.)

II. Reading Comprehension

Section A

Directions: For each blank in the following passage there are four words or phrases marked A, B, C and D. Fill in each blank with the word or phrase that best fits the context.

When is anger justified?

Anger is a complicated emotion. But is it ever morally right to be angry? And if so, when? One of the most foundational understandings of __21__ comes from the Greek philosopher Aristotle. In his model, there's a sweet spot for our actions and emotional reactions, and it's up to you to develop practical wisdom about when you should feel what and how strongly to feel it.

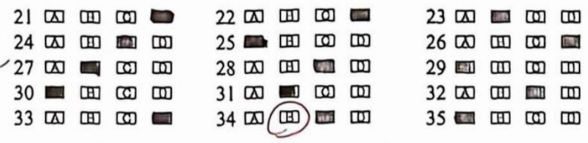
__22__, let's say you're going to sleep early because you have an important meeting tomorrow and your neighbor just started playing loud music. If you can't sleep, you might ruin your meeting, so feeling angry is definitely __23__. But how much anger should you feel? And what actions, if any, should you take? To answer these questions, Aristotle would need to know more details. Have

II. Reading Comprehension

Section A

Cloze

14



you __24__ talked to your neighbor about this issue? Is it a reasonable time to be playing music? Is your neighbor trying to __25__ you, or are they just enjoying their evening?

Relying on practical wisdom in Aristotle's case-by-case approach makes a lot of sense for handling __26__ conflicts. But what about when there's no one to __27__ for your anger? Imagine a tornado completely destroys your house while your neighbor's home is __28__. No amount of anger can undo the disaster, and there isn't really a suitable __29__ for your frustration.

Although it's hard for us to control our anger, there might be something we can learn from it. Philosopher PF Strawson's theory suggests that experiencing anger is a natural part of human psychology that helps us communicate blame and hold each other __30__. In this model, anger can be an important part of letting us know when something immoral is happening, so __31__ it would harm our social lives and moral communities. But finding the right response to those psychological alarm bells can be __32__. For instance, if you were supervising cruel, disrespectful young children, it might be natural to feel anger, but it would be __33__ to treat their moral mistakes like those of adults.

accountable

So when should you __34__ anger? And can it ever help change things for the better? Let's imagine your community is experiencing serious health issues because a nearby factory is __35__ polluting the water supply. In unjust situations like this, it could be a moral mistake to *suppress* (压制) your anger, instead of channeling it into positive action.

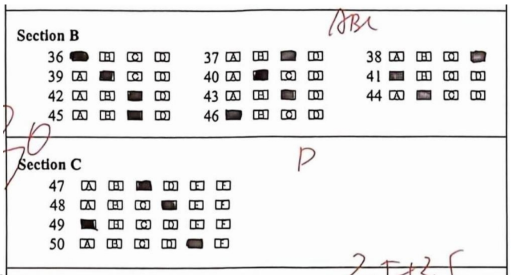
D	21. A. motion	B. passion	C. urge	D. anger
D	22. A. In conclusion	B. However	C. What's more	D. For example
B	23. A. exceptional	B. understandable	C. useless	D. tragic
C	24. A. remotely	B. reluctantly	C. previously	D. ultimately
A	25. A. upset	B. conquer	C. imitate	D. motivate
D	26. A. commercial	B. domestic	C. cultural	D. interpersonal 人际冲突
B	27. A. consult	B. blame	C. reject	D. hide
C	28. A. discovered	B. locked	C. untouched	D. exploded
A	29. A. target	B. boundary	C. position	D. reason
A	30. A. accountable	B. adorable	C. memorable	D. sustainable
B	31. A. expressing	B. removing	C. releasing	D. following → 结合下文
C	32. A. simple	B. dangerous	C. tricky	D. sufficient
D	33. A. mature	B. wise	C. easy	D. wrong
B ~~D~~	34. A. bring down	B. act on	C. bottle up	D. hold back
A	35. A. illegally	B. remotely	C. steadily	D. inevitably

Section B

Directions: Read the following three passages. Each passage is followed by several questions or unfinished statements. For each of them there are four choices marked A, B, C and D. Choose the one that fits best according to the information given in the passage you have just read.

(A)

Caroline Robbins knew that the first day of school was very, very important. Why? Because that was the day when you chose where you would sit for the entire year. Caroline realized that in



some classes this choice would be made for her. The teacher would place students in *alphabetical* (按字母顺序的) order, meaning she would have to sit in front of Zach Rodgers yet again. Zach was attracted by Caroline, and would distract her from her work by passing notes and telling jokes to impress her. In general, being a Robbins was pretty good, but having to sit in front of Zach was definitely a **drawback**.

In her other classes, though, Caroline would be sure to choose just the right seat. Caroline liked to think of herself as one of the cool kids, but she also did well in school and liked learning. So, Caroline wanted to sit close to the cool kids, but not too close, or she would be more interested in talking than paying attention. She also knew that it was good to be friends with the smart kids, because they could help Caroline with her schoolwork. However, she didn't want to sit too close to the smart kids. Unfortunately, at Caroline's school, the smart kids and the cool kids were not the same kids.

Caroline had all of this in mind as she walked through the school's front door on the first day. She knew where her first class was, but she didn't want to be the first one there. If you were the first one there, you didn't have any control at all! Other people got to choose how close they sat to you, not the other way around. This simply would not do. So, she took her time walking down the hall, taking a minute to talk to her friend Alma, whom she hadn't seen for the entire summer.

At last, she walked through the door of her first class, and there it was, the perfect seat! Two seats away from Jasmine, the smartest girl in her entire grade, and just in front of Marc, who was very cool and totally cute.

There was a new guy to the left, which could be a risk, but how bad could a new guy be? So Caroline started toward the seat, being careful not to rush, when Marc's best friend Jason sat down in her chair! Sure, there were other seats, but no other perfect ones. Saddened but not discouraged, Caroline sat down in the second-best seat and immediately started planning for lunchtime, when she was determined to get the best seat in the cafeteria.

36. As used in paragraph 1, the word "**drawback**" most nearly means "__________".
A. disadvantage B. benefit C. virtue D. withdrawal

37. What can be inferred from paragraph 2?
A. Caroline was often seen as a cool kid, not a smart kid.
B. Caroline wanted to be the only smart kid in her school.
C. If Caroline was seen as a smart kid, no one would think she was cool.
D. Caroline's school was unusual in that smart kids were also cool.

38. According to the passage, Caroline makes sure not to get to class too early by ________.
A. taking time to fix her hair B. going to the wrong class
C. waiting outside the school gate D. stopping to talk to her friend

39. How did Caroline feel about the new guy sitting next to her desired seat?
A. She was thrilled to have a new friend to sit next to.
B. She was concerned he might be a disturbance.
C. She was relieved that he didn't take the perfect seat.
D. She was disappointed because he didn't look cool.

完全不用读完,排除清楚.

(B)

From the time human beings began to draw them, maps have helped people find their way around their environments. But maps can show us many things, not just which direction to go. They show the path of history, the earth's shape development of mathematics, and the progress of technology.

One of the oldest-known land maps is an almost 1,600-year-old tablet from Southwest Asia. The map shows a circle of land that is divided by a river and surrounded by ocean. Triangles on the map indicate eight different regions. And the map's text describes legendary beasts and heroes that were important to the ancient people of the area. Around AD 150, a Greek scientist drew north-south and east-west lines on a map. This addition applies mathematics to mapping and was an early attempt to show the earth's shape on a flat piece of paper. Maps gradually became much more detailed as new regions were explored and put down on paper. Also, mathematical and astronomical advances helped to perfect the world map to what we know and love today!

Now that you know a little about the early history of maps, lets learn some fun facts.

East at the Top

These days, most maps feature north at the top. However, during the Middle Ages, most maps had east at the top. This was done to point in the direction of the morning sun.

Puzzle Maps

When printed maps became available to the general population in the eighteenth and nineteenth centuries, not everyone could understand them. In fact, the first *jigsaw puzzles* (拼图游戏) were designed as practice maps for eighteenth-century geography classes!

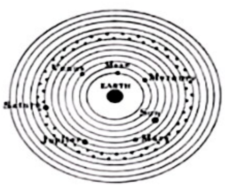
Fake Places

Mapmakers need to make sure that their work is not copied by others. To protect their work, many mapmakers add made-up towns or streets to their maps. Only the original mapmaker would know about the fake entry.

Modern Technology

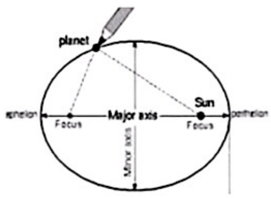
Today, digital maps and GPS technology have revolutionized the way we explore our world. With the touch of a screen, we can see our exact location, plan routes, and even view real-time traffic updates.

40. Which of the following pictures best shows the modern form of the Greek scientist's addition to the map?



A

B



C

D

B

41. Fake towns or streets were often added to maps __________.

A. so that it would be clear if someone copied the map

A.

B. to make the maps less boring and more accurate

C. as a tricky challenge for people to take on

D. to help people practice how to read complicated maps

C

42. What is the main purpose of the passage?

A. To show the development of mapping technology.

B. To explain the importance of maps throughout history.

C. To introduce the history of maps and some interesting facts.

D. To teach some map-making tricks and techniques.

(C)

Most taxi drivers need a smartphone to get to their destinations. But sharks, it seems, need nothing more than their own bodies—and Earth's *magnetic* (磁的) field. A new study suggests some sharks can read Earth's field like a map and use it to travel long distances with accuracy.

Since the 1970s, researchers have suspected that some fish can detect magnetic fields. But no one had shown that sharks use the fields to find their location or *navigate* (导航), partly because the animals aren't so easy to work with. It's one thing if you have a small fish, or a baby sea turtle, but when you work with sharks, you have to upscale everything.

Bryan Keller, an ecologist at Florida State University, and his colleagues decided to do just that. They lined a bedroom-size cage with wire and placed a small swimming pool in the center of the cage. By running an electrical current through the wiring, they could generate a custom magnetic field in the center of the pool. The team then collected 20 young bonnethead sharks—a species known to migrate hundreds of kilometers—off the Florida coast. They placed the sharks into the pool, one at a time, and let them swim freely under three different magnetic fields, applied in random succession. One field *mimicked* (模仿) Earth's natural field at the spot where the sharks were collected, while the others mimicked the fields at locations 600 kilometers north and 600 kilometers south of their homes.

They used software to track the sharks' responses, observing which direction in the tank they were trying to swim towards. When the young sharks were exposed to the magnetic field of the place they were captured, or 'home', they stayed put. But when subjected to the southern magnetic field, the sharks persistently changed their headings to swim north, as if trying to get back home. This suggests that the sharks were using the magnetic field to guide them, similar to how humans use GPS.

Surprisingly, the researchers found that the sharks didn't favor any direction when swimming under the northern field. Keller says this might be because they don't go north of their home location since there is only land there, and so they rarely have to find their way back south again. "This could support the theory that their ability to go back home is a learned behavior," he says. They might not know what to do in the northern field because "they've never been up there." says Keller.

Keller's research adds a significant piece to the still-incomplete puzzle of shark biology. Sharks have been declining at an alarming rate due to mostly overfishing and habitat change. Studying the life cycles and migration patterns of sharks can help us understand what areas to protect when managing marine spaces.

新细!!

43. Why is it difficult for researchers to prove that sharks can read Earth's field?

C

A. Sharks are too hard to follow and observe.
B. Sharks are not sensitive to magnetic fields.
C. Sharks are difficult to study in a laboratory setting.
D. Sharks are on the list of endangered species.

44. According to Keller, what might be the reason why sharks don't favor any direction when swimming under the northern magnetic field?

B

A. They don't like the climate in the north.
B. They've never been to the north of their home before.
C. They learned to do so when they were young.
D. The northern magnetic field was not strong enough.

45. From the passage, we can infer that Keller's research is significant because it ________.

C. ~~D~~

A. provides a new method for capturing sharks
B. supports the idea that sharks' migration patterns are random
C. adds crucial information to our understanding of shark biology
D. suggests that sharks should be protected from overfishing

46. What is the main idea of the passage?

A

A. Sharks use Earth's magnetic field to find their way on seas.
B. Researchers discovered a new way to study sharks in labs.
C. Shark populations are declining due to habitat change.
D. Sharks have a learned behavior of returning to their home.

Section C

Directions: Read the passage carefully. Fill in each blank with a proper sentence given in the box. Each sentence can be used only once. Note that there are two more sentences than you need.

~~A.~~ Unfortunately, our expectations can also work against us.
B. Believe it or not, there are limits to even the strongest placebo effect.
~~C.~~ This effect occurs when a patient is given a treatment that is able to improve their symptoms even though it lacks an active ingredient.
~~D.~~ Additionally, you probably associate pills in general with recovering from illness.
~~E.~~ If we can learn to make use of the power of positive thinking, perhaps one day we can even move beyond using traditional drugs to treat minor symptoms! 结尾展望.
F. And in routine medical practice, placebos are rarely used on purpose.

The power of *placebo* (安慰剂)

Have you ever taken a pill for a headache and felt instant relief even though there's no way the medicine could have taken effect so quickly? If so, you've personally experienced a medical phenomenon known as the placebo effect. ____47 C____ In the above situation regarding your headache, the active *ingredient* (成分) couldn't have been responsible for your reduced pain. So why did the symptoms improve?

This effect.

Scientists don't completely understand the mechanisms behind the effect, but they have determined that an individual's conditioning and expectations likely play a major role. With your headache, for example, you expected the painkiller to work because pills have relieved your headaches in the past. ____48____ In fact, the latter means a placebo can still work even when someone knows it's not real medicine! Other factors that can influence how well a treatment works include being told it's effective by a doctor, your doctor's body language or tone of voice, and the knowledge that you're taking action to solve the problem.

联想 D

against.

____49____ If you believe a drug will be ineffective at relieving your symptoms, for instance, it is less likely to work. Even worse, if you expect to suffer side effects from your medication, you probably will. This puts doctors in a tricky position—they're required by law to inform patients of risks, yet doing so could negatively affect recovery. The best way to avoid this undesirable effect, according to some experts, is for doctors to phrase dangers in the most encouraging way possible or ask people if they're willing to remain unaware of minor side effects.

A

The power of the placebo effect opens up an exciting opportunity to explore new avenues. Though we may not fully grasp how the placebo effects work, one thing is clear: our minds have an amazing ability to shape not only our thoughts but also our physical health. ____50____

E 结尾，总结

III. Summary Writing

Directions: Read the following passage. Summarize the main idea and the main point(s) of the passage in no more than 60 words. Use your own words as far as possible.

51.

Why we hate phone calls

Suddenly a sound rings out, stopping you in your tracks. Panicking, you search for where it could be coming from. It's your phone, and if you're like a quarter of 18 to 30-year-olds in a recent British study, you probably won't answer it. The same study found that 70% of the people in this age group prefer text messages to phone calls. Why do so many young people hate phone calls?

While previous generations grew up using landlines to talk to their friends, smartphone-equipped younger people have grown up used to using text messages for social conversations. There's less pressure with texting. You can read and respond to messages on your own schedule, and you can take time to think about what you want to say rather than being put on the spot during a phone call. Besides, when you can craft a reply free of interruption, you have greater control over your contributions to a conversation. Communication like phone calls can lead people to feel a loss of control and the corresponding anxiety. Many young people report associating phone calls, particularly those without prior warning, with bad news.

This means that new social codes are being established. Many people will now text someone to see if they're available to take a phone call. If someone doesn't feel able to sum something up in a few short messages, they might leave a long voice note.

But this doesn't mean that anxiety around communication has been stopped. Evidence suggests that texting can also cause anxiety. Many times, texting anxiety comes from frustration, fear, and worry over the reactions of other people. For instance, a short response to your long message might be interpreted as a cold shrug and many people report tensions from being "left on read"—when you know someone has read your message, but they don't, or won't, reply.

IV. Translation

Directions: Translate the following sentences into English, using the words given in the brackets.

52. 房间里堆满了主人从世界各地收来的古董。(fill)
53. 与人相处时，我们要学会换位思考并体谅他人的难处。(considerate)
54. 尽管加速通过黄灯能省点时间，但这种行为很可能引发严重的交通事故。(While)
55. 这场演唱会的舞台设计以创新和高科技元素而闻名，给观众带来无与伦比的视听感受。(experience)

V. Guided Writing

Directions: Write an English composition in 120-150 words according to the instructions given below.

56. 你是明启中学高三学生王华，你的好友李明来信邀请你和他一起创办一个公众号（a public account），通过文字、图片、视频等多种形式向国内外读者推广和传播中国文化。请你给他回信，内容须包括：

(1) 你对该想法的支持和赞赏；

(2) 你对合作的期待和对分工的建议。

2024 学年第一学期徐汇区学习能力诊断卷

高三英语　　答案

I. Grammar and vocabulary

1. which　2. produces　3. to manage/to be managed　4. are/were credited　5. fed
6. being given/given　7. must　8. if / when　9. what　10. themselves
11-20　BIAKH DCFEJ

II. Reading comprehension

Section A　21-35　DDBCA　DBCAA　BCDBA
Section B　36-39　ACDB　40-42 BAC　43-46 CBCA
Section C　47-50　CDAE

III. Summary Writing

51. Study found that young people nowadays prefer text messages to phone calls because their is less pressure and they can have more control over the communication, which reduces anxiety. New social code involves texting to check availability before calling or leaving a voice message. However, texting can create new anxieties about other people's reactions.

IV. Translation

52. The room is filled with antiques collected by the owner from all over the world.
53. When we get along with others, we should learn to put ourselves in others' shoes and be considerate of the difficulties of others.
54. While speeding through a yellow light might save a little time, this behavior is likely to result in a serious traffic accident.
55. The stage design of the concert is known for its innovation and high-tech elements, bringing the audience unparalleled/unmatched /incomparable/unique audio-visual experience.

V. Guided Writing

Dear Li Ming,

I was thrilled to receive your letter proposing the idea of starting a public account dedicated to promoting Chinese culture to readers both at home and abroad through diverse mediums like text, images, and videos. Your enthusiasm for sharing our rich heritage is truly commendable, and I fully support this endeavor.

The collaboration sounds like an exciting opportunity not only to educate but also to foster a deeper appreciation for Chinese culture among a global audience. I'm eagerly looking forward to partnering with you on this journey, where we can combine our strengths and creativity to make a meaningful impact.

Regarding the division of labor, I suggest we leverage each other's skills. You have a knack for storytelling and a keen eye for visual aesthetics, so perhaps you could focus on content creation and visual design and selecting captivating images and videos. On the other hand, I'm more technically inclined and proficient in digital platforms; I'd be happy to take charge of the technical aspects, such as setting up and maintaining the account, managing subscriptions, and optimizing our content for better reach and engagement.

Let's meet soon to brainstorm further and solidify our plan. Together, I believe we can turn this vision into a reality that resonates with hearts and minds worldwide.

Warmest regards,
Wang Hua